

# Khouloud ABID | Curriculum Vitae

Adress: Messina,Italy

 Nationality: Tunisian  Date of Birth: 04/07/1995  
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## Affiliations:

- Istituto di Tecnologie Avanzate per l'Energia "Nicola Giordano" (ITAE), Salita S. Lucia sopra Contesse n. 5 98126 – Messina, Italy
- Department of Engineering, University of Messina, I-98166 Messina, Italy
- CNR IPCF Istituto per i Processi Chimico-Fisici, viale F. Stagno D'Alcontres 37, I-98156 Messina, Italy
- Laboratory of Dielectric and Photonic Materials, Faculty of Sciences of Sfax, Sfax University, Sfax 3018, Tunisia;

## Scientific Background

I am a researcher in physics and chemistry. I am interested in the synthesis of two-dimensional (2D) materials, plasmonic nanoparticles, and their hybrid/nanocomposites. I am interested in studying the features of these materials and their performance in the sensing field, especially plasmonic, electrochemical, optical, and fluorescent. I can work with Raman, PL, UV-Vis, FTIR, and AFM techniques. On the other hand, I am interested in developing materials for thermochemical energy storage applications. I can work with XRD, BET, SETARAM, TGA, and DSC techniques. Moreover, I am interested in the synthesis and development of new materials for the thermochemical energy storage.

## EDUCATION

**Diploma of doctorate in 'Engineering and Chemistry of Materials and Constructions' and Diploma of doctorate in Condensed physics specialized in nanomaterials.** **19/12/2019- 12/12/2022**

*Department of Engineering, University of Messina-Italy and Department of Physics, University of Sfax-Tunisia* ◦ Title: "Development of Two-Dimensional (2D) Sensing Materials".

- Supervisor: Prof. Giovanni Neri and Prof. Ramzi Maalej/ Co-Supervisor: Prof. Pietro Gucciardi and Dr. Antonino Foti
- Competence: PIERS, SERS, synthesis plasmonic nanoparticles, synthesis 2D materials, sensing, electroanalytical technique, Raman, PL, UV, FTIR, AFM, synthesis of 2D nanocomposites
- Final decision: very good with written jury congratulations (in Tunisia)/ lode (in Italy)
- Delivered date: January 2023.
- Diplom are delivered from Faculty of sciences of Sfax-University of Sfax-Tunisia and University of Messina-Italy

### Master in Physics

**15/09/2017 - 06/12/2019**

*Dipartment of Physics, University of Sfax-Tunisia*

- Title: "PIERS and SERS Performance of MX2-Au Nanomaterials for MBA Detection".
- Supervisor: Prof. Ramzi Maalej/ Co-Supervisor: Prof. Nour Hamza Blekhir

- Competence: PIERS, SERS, plasmonic nanoparticles, 2D materials, sensing
- Final decision: 18.5/20 ( for the master defense with honors). 13.5/20 ( for the two years of the master) ◦
- Data di conseguimento: December 2019.
- Diplom is delivered from Faculty of sciences of Sfax-University of Sfax-Tunisia

**Diploma of Bachelor of Physics and Chemistry** **2014-2017**  
*Faculty of Sciences of Sfax-Tunisia*  
 ◦ Principle courses: Chemistry, physics, informatics, English and French languages

**Diploma of Baccalaureat (Mathematic)** **2014**  
*High School Maazoun, Sfax*  
 ◦ Final decision: 12.23/ 20; Near good  
 ◦ Date: 21/06/2014  
 ◦ Diploma delivered from Ministry of education of Tunisia

## PARTICIPANTS IN SCHOOLS

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- e-spark Summer school in electrochemistry  
**Date:** 18-24 September 2022. **Location:** Warsaw-Poland
- The International School on Programmable Smart Sensors based on Bio-compatible Nanocomposite Materials (NanoSENS)- Online **Date:**  
 17-24 June 2021

## RESEARCH ACTIVITIES

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**Post Doctoral** **01/04/2025 – 31/03/2026**  
*Istituto di Tecnologie Avanzate per l'Energia "Nicola Giordano" (ITAE), Salita S. Lucia sopra Contesse n. 5  
 98126 – Messina, Italy*

**Post Doctoral** **01/02/2024 – 31/01/2025**  
*CNR IPCF Istituto per i Processi Chimico-Fisici, viale F. Stagno D'Alcontres 37, I-98156 Messina, Italy*

**Post Doctoral** **01/01/2023 – 31/01/2024**  
*Department of Engineering, University of Messina-Italy*

## PROJECTS PARTICIPATION

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- Title of project: PRIN 2022 FLASH 2D  
 Role: my research focuses on nanomaterials, especially on transition metal dichalcogenides "TMDCs" (MX<sub>2</sub>; MoS<sub>2</sub>, WS<sub>2</sub>, MoSe<sub>2</sub>..The synthesis of these nanosheets is done through liquid-phase exfoliation (LPE) followed by liquid cascade centrifugation (LCC). The features of nanosheets (NS) were investigated by atomic force microscopy (AFM) technique to provide high-resolution imaging, determining the thickness and analyzing the defects. In addition to AFM, the MX<sub>2</sub> NS will be studied using tip-enhanced spectroscopies (TEPL and TERS).  
 Period: From 01/01/2023 to 31/12/2023.

◦ Title of project: TEcnologie innovative per il controllo, il moniToraggio e la sicurezza in mare (TETI)  
Role: Development of electrochemical sensors for the determination of contaminants in seawater Period:  
From 01/01/2023 to 31/12/2023.

◦ Title of project: SAMOTHRACE

Role: Development and characterization of a plasmonic sensor based on nanomaterials made of two-dimensional transition metal dichalcogenides in combination with noble metal nanoparticles for the detection of pollutants and toxins at low concentrations for applications of environmental interest. Period: From 01/01/2023 to 30/06/2023.

## PUBLICATIONS

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◦ **December 2025: " Electrochemical Detection of Cadmium Ions (Cd<sup>2+</sup>) at Low Concentration by a modified 2D-MoS<sub>2</sub>/SPCE"**

**Khouloud Abid**, Antonino Foti, Pietro Giuseppe Gucciardi, and Giovanni Neri. FlatChem **Under Revision**

◦ **April 2025: " Electrochemical Determination of Heavy Metals by a Graphene Quantum Dots/Porphyrin-Based Supramolecular System**

Mariachiara Sarà, **Khouloud Abid**, Pietro Giuseppe Gucciardi, Luigi Monsù Scolaro, and Giovanni Neri. Synthetic Metals

DOI: [10.1016/j.synthmet.2024.117824](https://doi.org/10.1016/j.synthmet.2024.117824)

◦ **December 2024: " Novel gold nanoparticles-Schiff base electrochemical sensor for the determination of lead (II) ions in biological samples Zahra Akbari, Khouloud Abid, Daniela Iannazzo, Morteza Montazerzohori, and Giovanni Neri. Bioelectrochemistry**

(2024), DOI: [10.1016/j.bioelechem.2024.108886](https://doi.org/10.1016/j.bioelechem.2024.108886)

◦ **September 2024: " Lead ions (Pb<sup>2+</sup>) Electrochemical Sensors Based on Novel Schiff Base Ligands**

Zahra Akbari, **Khouloud Abid**, Daniela Iannazzo, Morteza Montazerzohori, Enza Fazio, Fortunato Neri, Carmelo Corsaro, and Giovanni Neri. Environmental Science: Water Research & Technology (2024), DOI: [10.1039/D4EW00485J](https://doi.org/10.1039/D4EW00485J)

◦ **May 2024: " Graphene quantum dots (GQDs)-modified screen-printed electrode for the determination of cannabidiol (CBD) in hemp seeds flour"**

**K. Abid**, A. Ferlazzo, G. Neri FlatChem (2024), DOI: [10.1016/j.flatc.2024.100673](https://doi.org/10.1016/j.flatc.2024.100673)

◦ **October 2023: " A study of Screen-Printed Electrodes Modified with MoSe<sub>2</sub> and AuNPs-MoSe<sub>2</sub> Nanosheets for Dopamine Sensing"**

**Abid, K.**; Foti, A.; Khaskhoussi, A.; Celesti, C.; D'Andrea, C.; Polykretis, P.; Matteini, P.; Iannazzo, D.; Maalej, R.; Gucciardi, P. G.; Neri, G. *Electrochimica Acta* **2023**, 143371. DOI : [10.1016/j.electacta.2023.143371](https://doi.org/10.1016/j.electacta.2023.143371)

◦ **July 2023: " A sensitive and selective non-enzymatic dopamine sensor based on nanostructured Co<sub>3</sub>O<sub>4</sub>Fe<sub>2</sub>O<sub>3</sub> hetero-junctions. "**

Khan, M.; **Abid, K.**; Ferlazzo, A.; Bressi, V.; Espro, C.; Hussain, M.; Foti, A.; Gucciardi, P.; Neri, G. Chemosensors 2023, 11. DOI: [10.3390/chemosensors11070379](https://doi.org/10.3390/chemosensors11070379)

◦ **December 2022: "A novel 2D-GO@WS<sub>2</sub> electrochemical platform for the determination of thiram fungicide"**

**Abid, K.;** Iannazzo, D.; Celesti, C.; Khaskhoussi, A.; Foti, A.; Maalej, R.; Gucciardi, P. G.; Neri, G. *Journal of Environmental Sciences* 2024, 136, 226–236. DOI: [10.1016/j.jes.2022.11.018](https://doi.org/10.1016/j.jes.2022.11.018). ◦ **November 2022:**”

**Electrochemical and Sensing properties of AuNps-2D-MoS2/SPCE for Folic Acid Determination”**

**Abid, K.;** Zribi, R.; Maalej, R.; Foti, A.; Khaskhoussi, A.; Gucciardi, P. G.; Neri, G.

*FlatChem* 2022, 36, 100433. DOI: [10.1016/j.flatc.2022.100433](https://doi.org/10.1016/j.flatc.2022.100433).

◦ **August 2022: Chapter “Properties of 2D materials “** in Graphene and 2D Materials in Heat Transfer: Fundamentals and Applications

**Abid, K.;** Belkhir, N. H.; Wali, R.; Maalej, R

◦ **August 2020: “Photoinduced Enhanced Raman Spectroscopy with Hybrid Au@WS2 Nanosheets”**

**Abid, K.;** Belkhir, N. H.; Jaber, S. B.; Zribi, R.; Donato, M. G.; Di Marco, G.; Gucciardi, P. G.; Neri, G.; Maalej, R.

*J. Phys. Chem. C* 2020, 124 (37), 20350–20358. DOI: [10.1021/acs.jpcc.0c04664](https://doi.org/10.1021/acs.jpcc.0c04664)

## CONFERENCES, WORKSHOPS AND SEMINARS

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◦ **29-01 Sep-Oct 2025:** First International summer school for TCM friends academy in energy storage- Ljubljana, Slovenia

Title: “K<sub>2</sub>CO<sub>3</sub> composite TCM for low-medium temperature

Authors: **Khouloud Abid**<sup>1</sup> Antonio Fotia<sup>1</sup>, Roberta De Salvo<sup>1</sup>, Vincenza Brancato<sup>1</sup>, and Andrea Frazzica<sup>1</sup> Oral presentation.

Role: presenter

◦ **23-25 September 2024:** First Arab-German Plasmonic Sensor Workshop- Cairo-Egypt

Title: “Photo-induced enhanced Raman spectroscopy (PIERS) based on 2D materials”

Authors: **Khouloud Abid**<sup>1</sup> Antonino Foti<sup>1</sup>, Pietro Giuseppe Gucciardi<sup>1</sup> Oral

presentation.

Role: presenter

◦ **10-12 July 2024:** Plasmonica Workshop, Messina-Italy

Title: “Photo-Induced Enhanced Raman Spectroscopy Beyond Two-Dimensional Materials”

Authors: **Khouloud Abid**<sup>1,2,3\*</sup>, Nour H. Belkhir<sup>3,4</sup>, Sultan B. Jaber<sup>5</sup>, Rayhane Zribi<sup>1,2</sup>, Maria Grazia Donato<sup>1</sup>, G. Di Marco<sup>1</sup>, Antonino Foti<sup>1</sup>, Pietro G. Gucciardi<sup>1</sup>, Giovanni Neri, and Ramzi Maalej<sup>3</sup> Oral presentation.

Role: presenter

◦ **25-28 June 2024:** Graphene 2024 conference, Madrid-Spain

Title: “PTwo-Dimensional Materials (2D) from Synthesis to High-Performance Electrochemical Sensors”

Authors: **Khouloud Abid**<sup>1,2</sup>, Antonino Foti<sup>1</sup>, Ramzi Maalej<sup>3</sup>, Pietro G. Gucciardi<sup>1</sup>, and Giovanni Neri<sup>2</sup> Oral presentation.

Role: presenter

◦ **3-7 June 2024:** The International School on Plasmonics and Nano-Optics. Lake-Como, Italy

Title: “Photo-Induced Raman Spectroscopy (PIERS) on AuNps-WS<sub>2</sub> Nanohybrid”

Authors: **Khouloud Abid**<sup>1,2,3\*</sup>, Nour H. Belkhir<sup>3,4</sup>, Sultan B. Jaber<sup>5</sup>, Rayhane Zribi<sup>1,2</sup>, Maria Grazia Donato<sup>1</sup>, G. Di Marco<sup>1</sup>, Antonino Foti<sup>1</sup>, Pietro G. Gucciardi<sup>1</sup>, Giovanni Neri, and Ramzi Maalej<sup>3</sup> Oral presentation.

Role: presenter

◦ **29-30 May 2023:** 7th International Conference on Material Science and Engineering- Barcelona-Spain

Title: “Novel sensitive platform toward thiram pesticide based on GO:WS2/SPCE”

Authors: **Khoulood Abid**<sup>1,2,3</sup>, Daniela Iannazzo<sup>1</sup>, Consuelo Celesti<sup>1</sup>, Amani Khaskhoussi<sup>1</sup>, Antonino Foti<sup>2</sup>, Ramzi Maalej<sup>3</sup>, Pietro Giuseppe Gucciardi<sup>2</sup>, Giovanni Neri<sup>1\*</sup> Oral presentation.

Role: presenter

- **22-25 May 2022:** 7th International Conference on Bio-Sensing Technology- Sitges-Spain

Title: "A New Electrochemical Like-Immunosensor Based on Au@MoS<sub>2</sub> hybrid for The Detection of Vitamin B<sub>9</sub>"

Authors: **Khoulood Abid**<sup>1,2,3,\*</sup>, Ramzi Maâlej<sup>1</sup>, Antonino Foti<sup>3</sup>, Pietro G. Gucciardi<sup>3</sup>, Giovanni Neri<sup>2</sup>

Oral presentation

Role: presenter

- **10-11 February 2022:** RSC Chemical Nanoscience and Nanotechnology Early Careers Virtual Meeting - Online

Title: "PIERS Hybrid Au@WS<sub>2</sub> Nanosheets for Sensitive Detection of 4-Mercaptobenzoic Acid"

Authors: **K.Abid**<sup>1</sup>, N.Hamza Belkhir<sup>+</sup>, R.Zribi<sup>3,4</sup>, S.Jaber<sup>5</sup>, P.Gucciardi<sup>6</sup>, G.Neri<sup>3,4</sup>, R.Maalej<sup>1</sup>

Poster presentation

Role: presenter

- **10-11 February 2022:** AISEM - Online

Title: "New Electrochemical Platform Au@MoS<sub>2</sub> for Sensing Aim at Low Concentrations"

Authors: **Khoulood Abid**<sup>1,2,3,4</sup>, Giovanni Neri<sup>2,3</sup>, Pietro G. Gucciardi<sup>4</sup>, Ramzi Maâlej<sup>1</sup>

Poster presentation

Role: presenter

- **4-7 February 2021:** AMEG - Online

Title: "New PIERS Substrate for The Detection at Low Traces Based on WS<sub>2</sub> Nanosheets and Gold Nanoparticles"

Authors: **K.Abid**<sup>1\*</sup>, N.Hamza Belkhir<sup>1,2</sup>, R.Zribi<sup>1,3,4</sup>, S.Jaber<sup>5</sup>, P.Gucciardi<sup>6</sup>, G.Neri<sup>3,4</sup>, R.Maalej<sup>1</sup>

Oral presentation

Role: presenter

- **07-09 December 2020 :** 6th Nano Boston Conference -Online

Title: "The efficiency of Photo-Induced Surface Enhanced Raman Spectroscopy (PIERS) WS<sub>2</sub>-AuNPs sensor in the detection of MBA residues"

Authors: **Khoulood Abid**<sup>1\*</sup>, Nour H.Belkhir<sup>1,2</sup>, Sultan B.Jaber<sup>3,4</sup>, Rayhane Zribi<sup>1,5,6</sup>, Pietro Gucciardi<sup>7</sup>, Giovanni Neri<sup>5,6</sup>, Ramzi Maâlej<sup>1</sup> Poster presentation:

Role: presenter

- **29-31 October, 2019:** 4th International Conference of Engineering Sciences for Biology and Medicine (ESBM)

Title: "The SERS tool performance on MoS<sub>2</sub>-AuNPs"

Authors: **K.Abid**<sup>1</sup>, N.Hamza Belkhir<sup>1,2</sup>, S.Jaber<sup>3</sup>, R.Zribi<sup>1,4</sup>, P.Gucciardi<sup>4</sup>, G.Neri<sup>5</sup>, R.Maalej<sup>1</sup> Oral presentation:

Role: presenter

## LECTURES

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- Seminar at Department of Engineering of University of Messina-Italy.

Title: "Vitamin B<sub>9</sub> Electrochemical Determination Based on MX<sub>2</sub>-Au NPs"

Event name: Doctorate course (XXXV cycle) Date:

5<sup>th</sup> May 2022

- Online teaching physics for high-school pupils on YouTube Channel Link:  
[https://www.youtube.com/watch?v=tBcZpnTdfx4&t=7s&ab\\_channel=NokhbaTN](https://www.youtube.com/watch?v=tBcZpnTdfx4&t=7s&ab_channel=NokhbaTN)

## OTHER ACTIVITIES

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- Supervision of three students in 2021-2022 and 2022-2023
- **November 2022:** Editor member of American Journal of Chemical and Biochemical Engineering(AJCBE)

## PERSONAL EXPERIENCES

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2018-2019: Giving courses on mathematics and English for orphan kids in Sfax charity    Developing Website for my Family clothes brand

## COMPETENCES

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### DIGITAL AND SOFTWARE COMPETENCES

Office365, Origin, Nova, DropView, VESTA, Chemdraw, ImageJ, LabSpec6, Gwydion, FPS, adobe photoshop. Average knowledge in html language, MATLAB and python

### LANGUAGES

**Arabic:** Native language; **French:** B2 in speaking and B1 in writing; **English:** B2 B1 in speaking and B2 in writing; **Italian:** A2 in speaking and A1 in writing ; **Turkish:** A1 in speaking and A1 in writing.